

Week 8: Fractions I

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

Fractions appear whenever something is divided into parts — sharing food, measuring ingredients, or dividing money among family members. This week introduces the language and basic types of fractions.

What is a Fraction?

A fraction represents a part of a whole. It is written as **numerator/denominator** (e.g., $\frac{3}{4}$), where:

- **Numerator** = the top number (how many parts we have)
- **Denominator** = the bottom number (how many equal parts the whole is divided into)

Types of Fractions

1. **Proper fraction** — numerator is smaller than denominator (e.g., $\frac{3}{5}$, $\frac{7}{8}$)
2. **Improper fraction** — numerator is equal to or larger than denominator (e.g., $\frac{7}{4}$, $\frac{9}{9}$)
3. **Mixed number** — a whole number combined with a proper fraction (e.g., $2\frac{1}{3}$)

Converting Improper Fractions ↔ Mixed Numbers

Improper → Mixed:

Example 1: $11/4 \rightarrow 11 \div 4 = 2$ remainder 3 $\rightarrow 2\frac{3}{4}$

Example 2: $17/5 \rightarrow 17 \div 5 = 3$ remainder 2 $\rightarrow 3\frac{2}{5}$

Mixed → Improper:

Example 1: $3\frac{2}{5} \rightarrow (3 \times 5 + 2)/5 = 17/5$

Example 2: $4\frac{3}{4} \rightarrow (4 \times 4 + 3)/4 = 19/4$

Simplifying Fractions

To simplify a fraction, divide the numerator and denominator by their **HCF**.

Example 1: Simplify $18/24$.

HCF of 18 and 24 = 6

$18 \div 6$ over $24 \div 6 = \frac{3}{4}$

Example 2: Simplify $42/56$.

HCF of 42 and 56 = 14

$42 \div 14$ over $56 \div 14 = \frac{3}{4}$

Real-Life Application

If a mother divides a bag of rice into 4 equal parts and gives 3 parts to a family, she has given away $\frac{3}{4}$ of the bag — a proper fraction. If a recipe uses 9 half-cups of flour, that is

$\frac{9}{2}$ cups (an improper fraction), which can be rewritten as $4\frac{1}{2}$ cups (a mixed number) for easier measuring.

Vocabulary

Word	Meaning
Numerator	The top number of a fraction, showing how many parts are taken
Denominator	The bottom number of a fraction, showing how many equal parts make a whole
Simplify	To reduce a fraction to its lowest terms

Evaluation Questions

- 1 Classify: $\frac{2}{5}$, $\frac{9}{5}$, $4\frac{2}{3}$, $\frac{3}{3}$ (proper, improper, or mixed?)
- 2 Convert $\frac{17}{5}$ to a mixed number.
- 3 Convert $4\frac{3}{4}$ to an improper fraction.
- 4 Simplify $\frac{42}{56}$ to its lowest terms.
- 5 Explain the difference between the numerator and the denominator.
- 6 Convert $\frac{23}{6}$ to a mixed number.
- 7 Convert $5\frac{3}{8}$ to an improper fraction.
- 8 Simplify $\frac{36}{48}$ to its lowest terms.
- 9 A cake is cut into 8 equal slices, and 5 are eaten. Write this as a fraction and state its type.
- 10 Explain, using your own example, why $\frac{7}{7}$ is neither a proper nor a typical mixed number.